

SFM Technologies

SUMMARY

NeedHub is a location-aware and skill-aware request routing platform where users can post requests, receive matched helpers, express interest in helping, chat privately, complete requests, and rate each other. The system routes requests based on skills, location, request category, user reputation, and user activity.

PROJECTS

NeedHub – Dynamic Help Routing Platform | Technical Requirements Document (TRD), Version 1.0, Status: Draft

- Location-aware and skill-aware request routing platform
- Core modules include Authentication, User Profile, Skills, Request Management, Feed Routing Engine, Interest Management, Chat, Ratings, Notifications, and Admin
- Routing formula: 40% skill match, 30% distance, 15% activity, 10% reputation, 5% urgency
- Authentication includes JWT access token (15 minutes) and refresh token (30 days)
- Infrastructure includes AWS EC2, AWS RDS, AWS S3, CloudFront, and Nginx
- MVP scope includes Authentication, Profiles, Skills, Request Posting, Routing Engine, Interest Flow, Chat, Ratings, Notifications, and Admin Portal

TECHNICAL SKILLS

React

TypeScript

Tailwind CSS

Vite

Node.js

Express.js

PostgreSQL

AWS S3

JWT

Refresh Tokens

Firebase Cloud Messaging

Email Service

AWS EC2

AWS RDS

CloudFront

Nginx

bcrypt

SYSTEM MODULES

Authentication Module; User Profile Module; Skills Module; Request Management Module; Feed Routing Engine; Interest Management Module; Chat Module; Ratings Module; Notifications Module; Admin Module

API ENDPOINTS

Authentication: POST /auth/register, POST /auth/login, POST /auth/logout, POST /auth/forgot-password. Users: GET /users/profile, PUT /users/profile, GET /users/skills, PUT /users/skills. Requests: POST /requests, GET /requests, GET /requests/:id, PUT /requests/:id, DELETE /requests/:id. Interest: POST /requests/:id/interest, POST /interest/:id/accept,

POST /interest/:id/reject. Chat: GET /conversations, POST /messages, GET /messages. Ratings: POST /ratings, GET /ratings.

PERFORMANCE TARGETS

API Response <500ms; Feed Load <2 seconds; Chat Delivery <1 second; Image Upload <5 seconds; Search Results <2 seconds; System Availability 99.9%.

MVP RELEASE SCOPE

Included: Authentication, Profiles, Skills, Request Posting, Routing Engine, Interest Flow, Chat, Ratings, Notifications, Admin Portal. Excluded: Payments, Escrow, Subscriptions, Video Calls, Business Accounts, AI Agents, Marketplace Bidding, Advanced Analytics.

FUTURE RELEASES

V2: Payments, Escrow, Scheduling, Calendar, Background Checks, Premium Accounts. V3: AI Matching, AI Assistant, Voice Requests, Voice Search, Smart Recommendations. V4: Enterprise Accounts, Business Requests, API Ecosystem, Third-Party Integrations, White Label Platform.